RAJALAKSHMI ENGINEERING COLLEGE

Ramanan K S 251001207-2025-IT

R2

Started on	Friday, 14 August 2026, 2:27 PM
State	Finished
Completed on	Wednesday, 26 August 2026, 8:43 PM
Time taken	12 days 6 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1

Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2
3  void Factor(int num)
4  {
5      int i;
6      long count = 0;                          // declaration - not counted
```

```
7
8       for (i = 1; (count++, i <= num); (i++, count++))   // test counted every check,
9                                                            // increment counted every pass
10      {
11          if (num % i == 0)
12          {
13              count++;                                  // counted only when divisor found
14          }
15      }
16
17      printf("%ld\n", count);
18  }
19
20  int main(){
21      int n;
22      scanf("%d",&n);
23      Factor(n);
24      return 0;
25  }
```

	Input	Expected	Got	
✔	12	31	31	✔
✔	25	54	54	✔
✔	4	12	12	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course